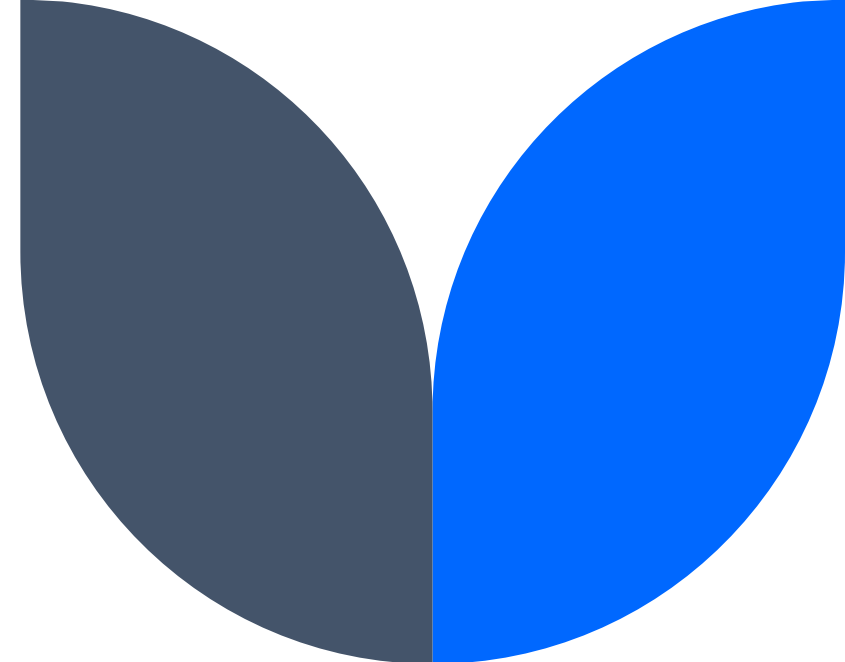
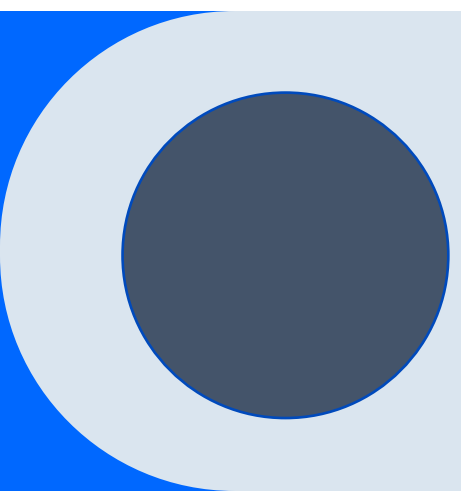




Dislocation Reductions in Austere Environments



Sarah Frances McClure, DO, FAWM, LP
Medical Director, Timberline Medics
4/18/2024





Agenda

Introduction

Shoulders

Patella

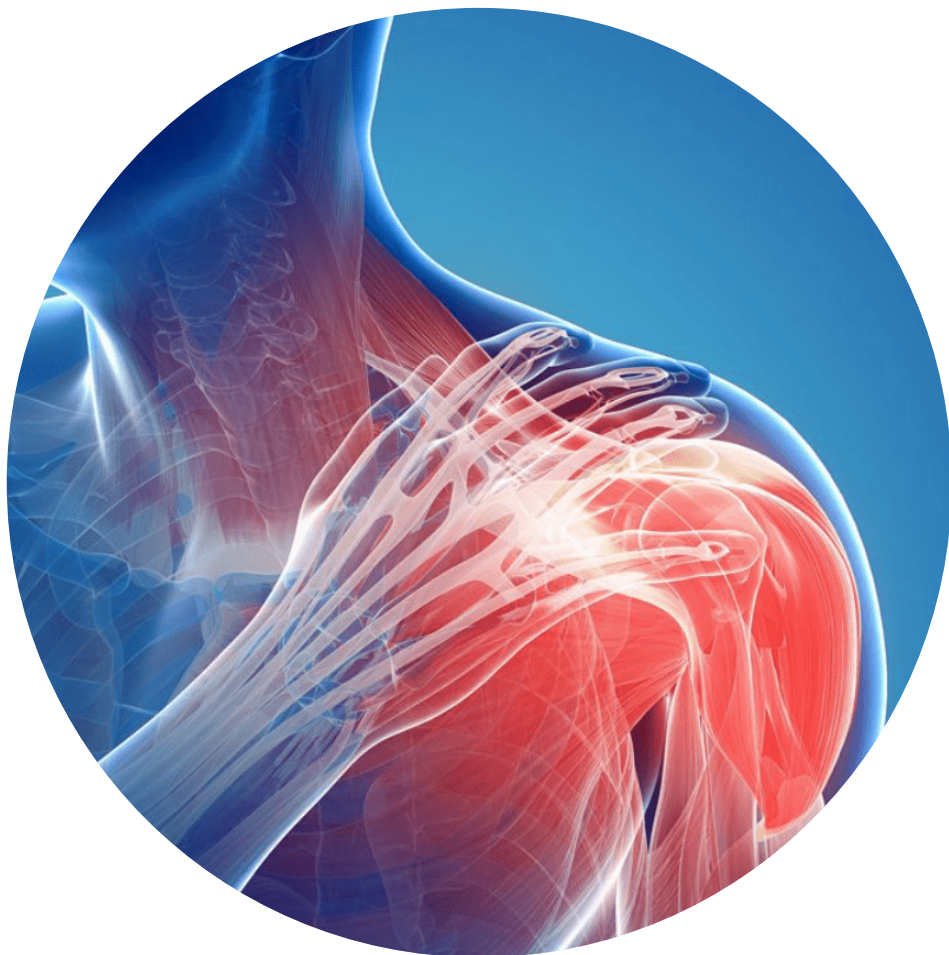
Fingers



Medical Director

Wildland Fire
Industrial Medicine
Event Medicine

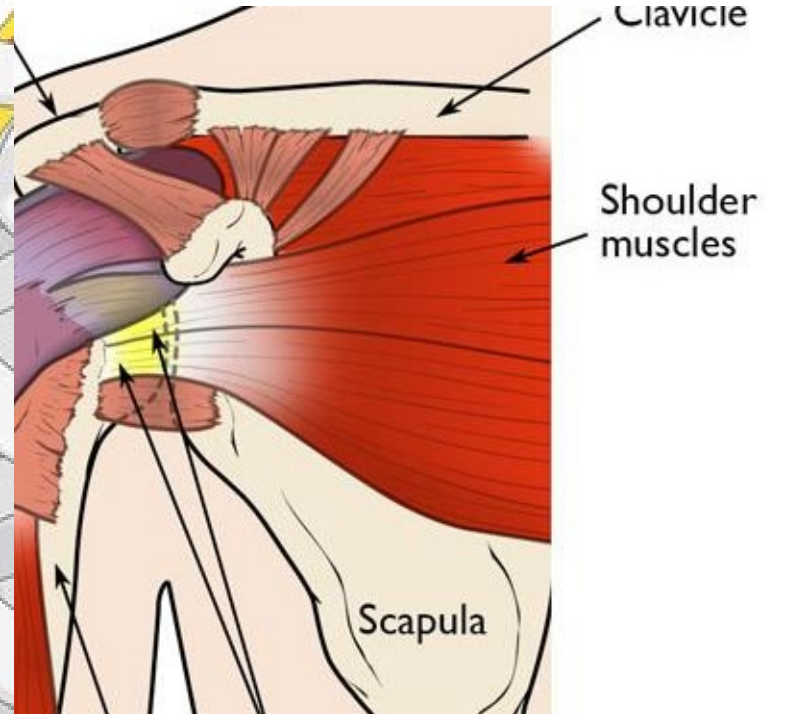
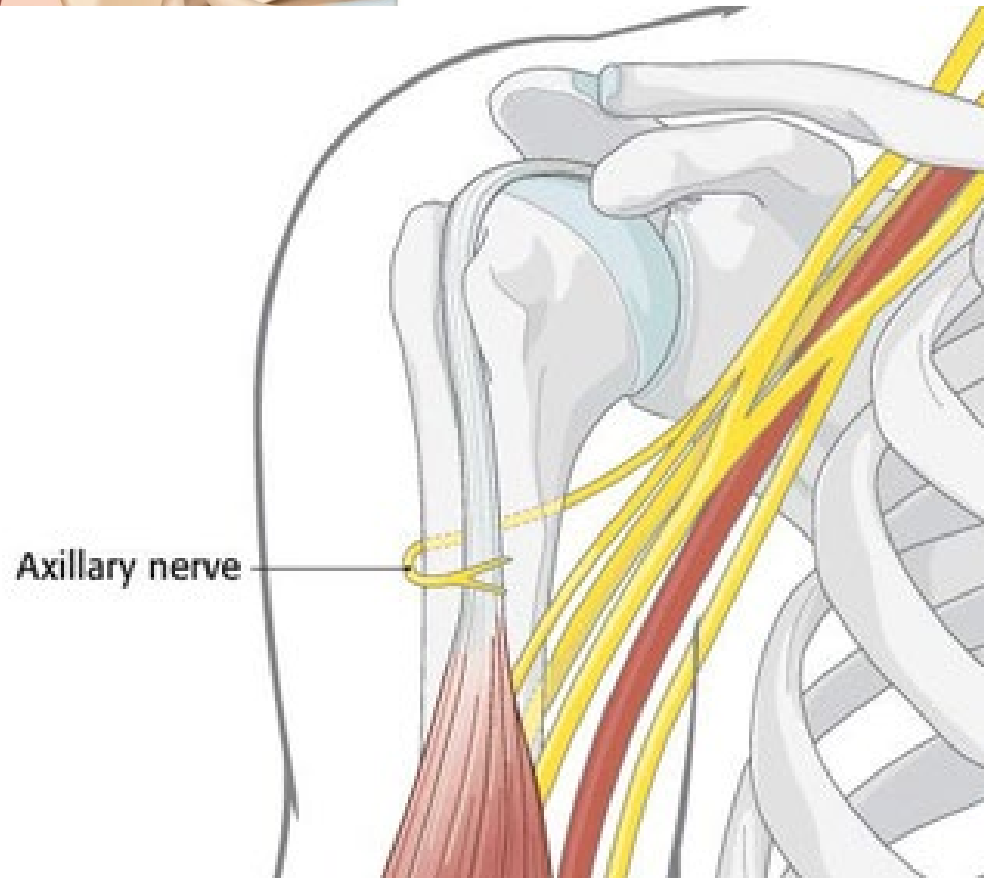
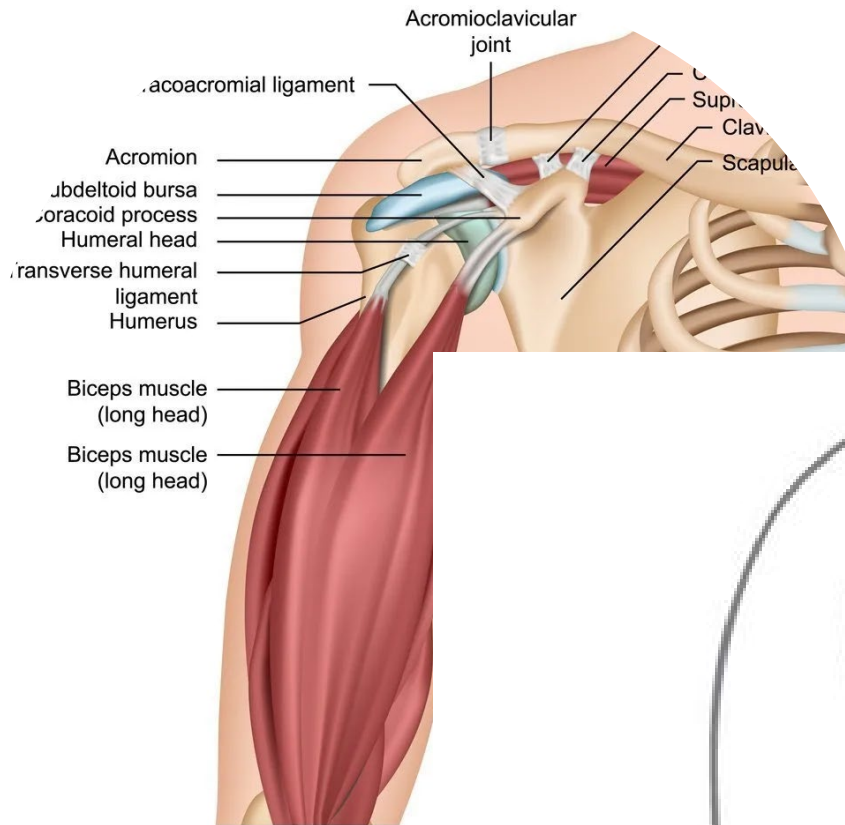




Shoulders

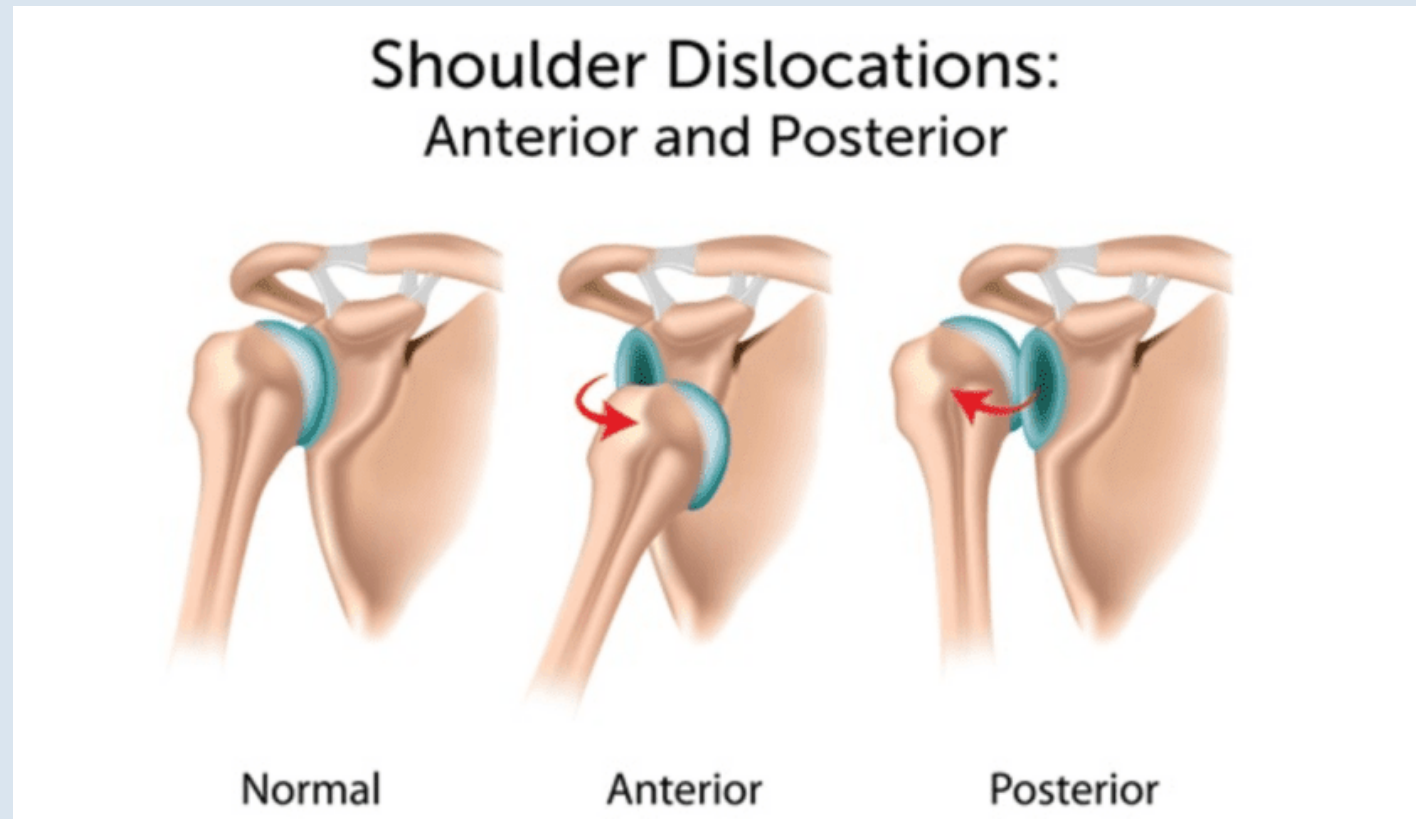


Shoulder Anatomy



Types of Shoulder Dislocations

- 90-98% anterior
 - Most FOOSH or MVC
- Posterior 5%
 - Post seizure
- Inferior 1%



Exam

Pain

Arm maintained in
abduction and
external rotation

Shoulder appears
“squared off”

Difficulty touching
affected arm to
contralateral shoulder





Reduction

Procedure

1. Pain Management
2. Pulse, Motor, Sensory Exam
3. Avoid pressure on the axillary nerve
4. Utilized one of the approved methods while coaching the patient to relax
5. Pulse, Motor, Sensory Exam
6. Sling

Discussion/Consent

Benefit

- Decrease risk of neurovascular damage
- Increase success of reduction without sedation or surgery

Risk

- Undiagnosed fracture
 - Hill-Sachs Lesion 40%
 - Bankart Lesion 10-20%
 - Bony Bankart Lesion
 - Rotator Cuff Tear
 - Pain
 - Neurovascular damage
 - Axillary nerve injury 14% (usually self-resolves)
 - Axillary artery rupture
 - Fracture
 - Unsuccessful attempt and requiring sedation or surgery
- 

Davos Technique



Scapula Rotation Technique



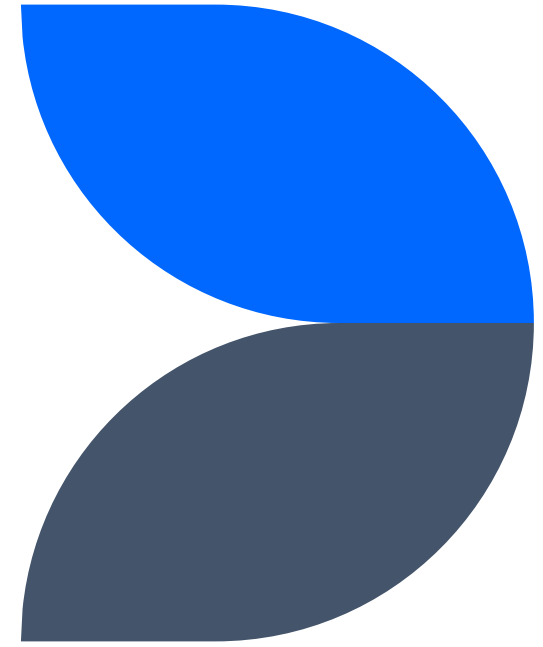
Cunningham Technique



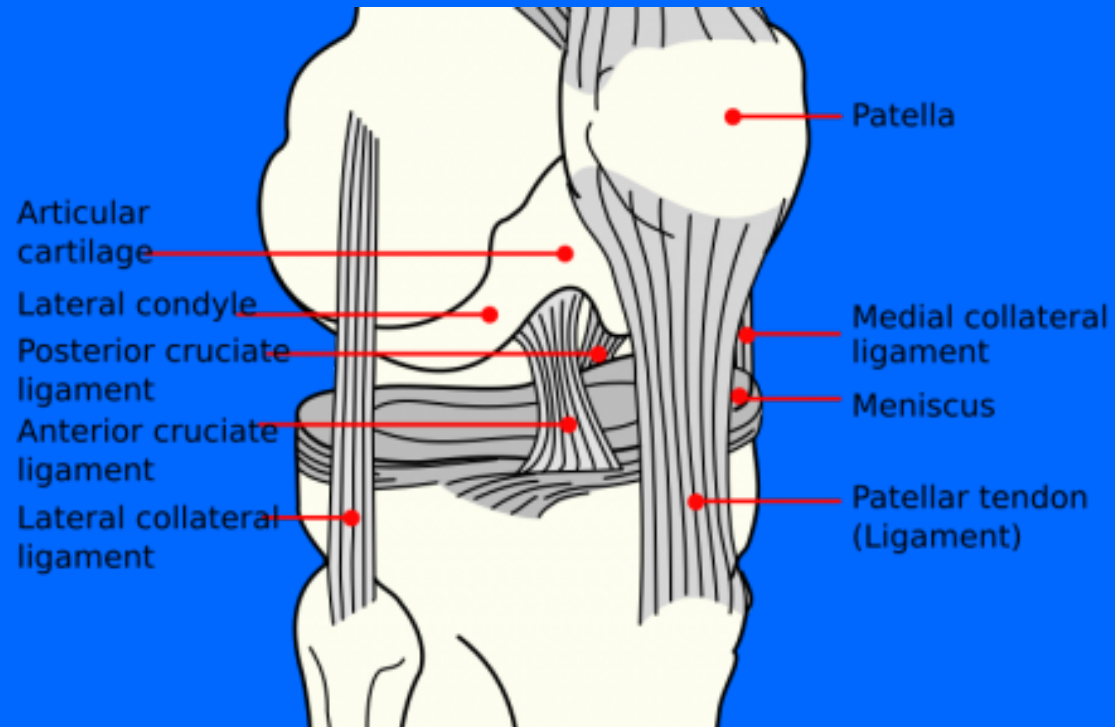
Spaso Technique



Patella Dislocation



Patella Anatomy



Exam/Reduction

1. Most patella will dislocation laterally
2. Perform the exam in full knee extension
3. Gently push the patella back into place
4. Immobilize knee
5. Provide pain medication (NSAIDS)

Discussion

Benefits

- Ability to walk out of the austere environment
- Pain reduction
- Reduce general harm

Risks

- Pain
- Undiagnosed fracture
 - Medial patellar facet
 - Lateral femoral condyle

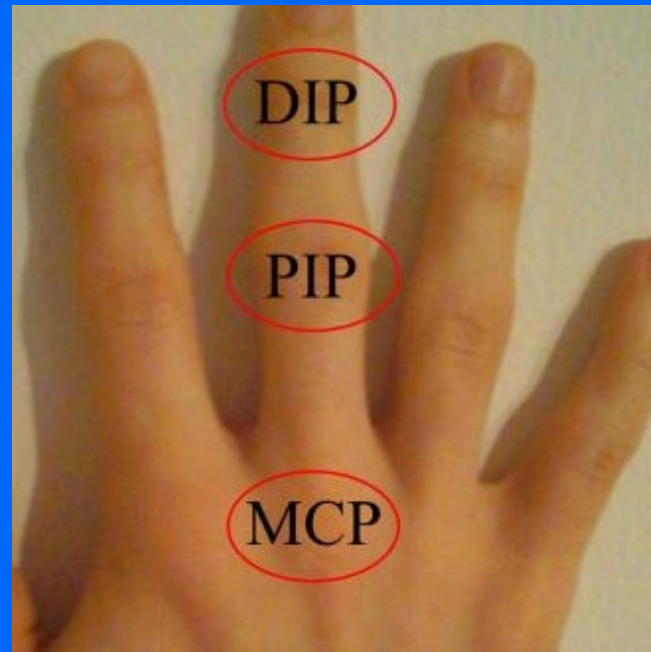
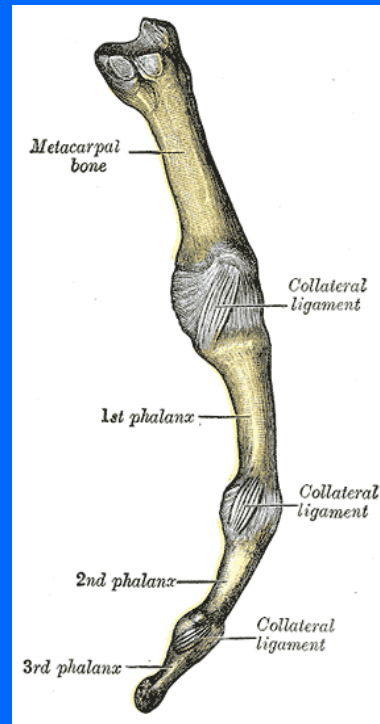
Patella Reduction



Finger Dislocation



Finger Anatomy



Exam/Reduction

1. Can be dislocated at 1 or multiple joints
2. Can be open or closed dislocations
3. May have skin color changes or decreased capillary refill
4. Can be dislocated at any of the joints
5. Grasp the finger and pull longitudinally and in direction of dislocated piece to disengage from anatomically correct bone
6. Flex or extend (opposite of what was just done to disengage)
7. Splint!!

Discussion

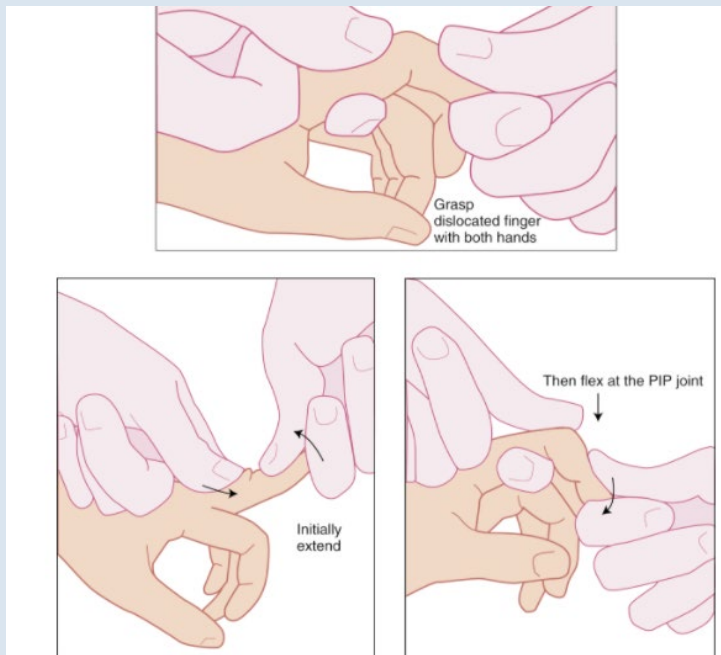
Benefits

- Increase blood flow
- Pain reduction
- Maintain anatomical position for better functional outcome

Risks

- Pain
- Neurovascular damage
- Undiagnosed fracture
- If open, then infection

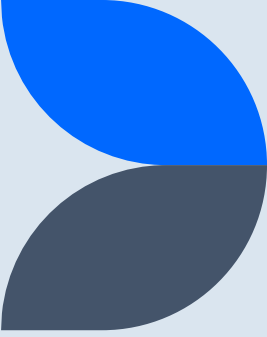
Finger Dislocation Reduction



Imaging

All reductions need imaging

All reductions should be transported to a medical facility for imaging



Thank you



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